

Proposed Schema for AY 25-26

PROPOSED SCHEMA FOR SEMESTER VII

Semester VII Teaching Scheme									
Course Type	Course Code	Course name	Teaching scheme (Contact Hours)			Credits assigned			Total
			Th	Pr	Tu	Th	Pr		
PCC	NITPC71	Blockchain Technology	2	2	-	2	1	-	3
PCC	NITPC72	Big Data Analytics	2	2	-	2	1	-	3
PEC	NITPEC71	1. Data Science 2. Social Media Analytics	3	2	-	3	-	1	4
PEC	NITPEC72	1. Generative AI 2. Soft Computing	3	2	-	3	-	1	4
RM	NITRM72	RM	3	2	-	3	1	-	4
Project	NITPR71	Project	-	8	-	-	4	-	4
			13	18	0	13	7	2	22
Total Hours			31			Total Credits			22

Semester VII Examination Scheme									
Course Type	Course Code	Course Name	Theory				Term Work	Pract & oral	Total
			Internal Assessment		End Sem Exam	Exam Duration (in Hrs)			
			Mid Test	CA					
PCC	NITPC71	Blockchain Technology	20	20	60	2	25	25	150
PCC	NITPC72	Big Data Analytics	20	20	60	2	25	25	150
PEC	NITPEC71	1. Data Science 2. Social Media Analytics	20	20	60	2	25	-	125
PEC	NITPEC72	1. Generative AI 2. Soft Computing	20	20	60	2	25	-	125
RM	NITRM72	RM	20	20	60	2	25	-	125
Project	NITPR71	Project	-	-	-	-	50	50	100
TOTAL MARKS									775



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Program Electives

PEC1		PEC2	
Course Code	Course Name	Course Code	Course Name
NITPEC711	Data Science	NITPEC721	Generative AI
NITPEC712	Social Media Analytics	NITPEC722	Soft Computing

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COURSE NAME: **DATA SCIENCE**

Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/PR	Tut	Total
NITPEC711	Data Science (Theory)	03	---	---	03	—	---	03
	Data Science (Lab)	—	02	---	—	01	---	01

Examination Scheme							
Course Code	Course Name	Theory			Term Work	Practical & Oral	Total
		Internal Assessment		End Semester Exam			
		Mid-Term Test	Continuous Assessment				
NITPEC711	Data Science (Theory)	20	20	60	---	---	100
	Data Science (Lab)	—	—	—	25	---	25

Course Prerequisite: Python programming, Data Structures

Course Objectives:

1	To introduce the complete Data Science process and handling of structured and semi-structured data from multiple sources.
2	To develop the ability to perform statistical analysis and exploratory data analysis for understanding data characteristics.
3	To apply appropriate data visualization and validation methodologies for effective interpretation of data.
4	To understand and implement anomaly detection techniques for identifying unusual patterns in data.
5	To apply time series forecasting methods for analyzing and predicting temporal data.
6	To understand optimization concepts and explore real-world applications of Data Science with ethical considerations.

Course Outcomes:

After successful completion of the course students will be able to:

1	Acquire, clean, and transform data from structured and semi-structured sources using appropriate tools.
2	Perform descriptive, inferential statistical analysis and exploratory data analysis on real datasets.
3	Apply visualization and model validation techniques to interpret and present data insights effectively.
4	Detect outliers and anomalies using statistical and machine learning approaches.
5	Build and evaluate time series forecasting models for predictive analysis.
6	Apply optimization techniques and Data Science methods to solve real-world problems while addressing ethical and privacy issues.

Learning Outcomes:

1	Describe and apply the different methods for analysis(time and space complexity) of algorithms.
2	Implement advanced data structures like B/B+ Trees, Red-Black Trees etc..
3	Apply an appropriate algorithmic design approach (divide and conquer and greedy approach) to solve a given problem.
4	Apply the dynamic programming technique to solve a given problem.
5	Select and apply an appropriate pattern matching algorithm for a given application.

Syllabus

Sr. No.	Module	Detailed Contents	Hours
1	Introduction to Data Science	1.1 Introduction to Data Science; Data Science Process (CRISP-DM); Types of data – structured, semi-structured, unstructured (text, logs, JSON). 1.2 Motivation to use Data Science Techniques: Volume, Dimensions and Complexity; Data Science tasks and examples; Data sources – CSV, APIs, web data; Handling JSON, log files, and API data using Python. 1.3 Overview of Data Preparation and Modeling; Difference between Data Science, Data Analytics, and AI; Overview of Python ecosystem – Pandas, NumPy, Jupyter.	5
2	Data Exploration	2.1 Descriptive Statistics & EDA: Univariate exploration – Measures of Central Tendency (Arithmetic Mean, Weighted Mean, Median, Mode); Measures of Dispersion (Range, Quartile Deviation, IQR); Measures of Skewness (Karl Pearson's, Bowley's); Measures of Kurtosis. Multivariate exploration – Correlation analysis, Bivariate distribution, Covariance, Types of correlation, Karl Pearson's	10

		coefficient; Correlation heatmaps, pairplots; Handling missing values and outliers during EDA. 2.2 Inferential Statistics: Distributions (Normal, Poisson); Statistical inference and tests of significance; Hypothesis testing procedure; Tests in attributes; Test of significance of a single mean; Central Limit Theorem; Confidence Interval; Z-test, t-test; Type I & Type II errors; F-distribution and ANOVA; Practical interpretation of p-value in data science.	
3	Methodology and Data Visualization	3.1 Methodology: Overview of model building; Cross validation; K-fold cross validation; Leave-one-out; Bootstrapping; Model evaluation metrics for classification and regression in data science workflow. 3.2 Data Visualization: Univariate – Histogram, Quartile plot, Distribution chart. Multivariate – Scatter plot, Scatter matrix, Bubble chart, Density chart; Roadmap for data exploration; Visualization using Matplotlib, Seaborn, Plotly; Data storytelling, report preparation, and basic dashboard concepts.	6
4	Anomaly Detection	4.1 Outliers, Causes of outliers, Anomaly detection techniques; Outlier detection using statistical methods. 4.2 Distance-based outlier detection; Density-based outlier detection; Isolation Forest; Local Outlier Factor (LOF); Handling imbalanced datasets using SMOTE; Use cases – fraud detection, network intrusion.	6
5	Time Series Forecasting	5.1 Taxonomy of time series forecasting methods; Time series decomposition; Concept of stationarity and ADF test; Train-test split for time series. 5.2 Smoothing methods – Average method, Moving average; Time series using linear regression; ARIMA model; Introduction to Prophet model; Performance evaluation metrics – MAE, RMSE, MAPE, MASE.	8
6	Optimization Techniques and Applications of Data Science	6.1 Optimization: Global and local optima; Unconstrained and constrained optimization; Introduction to least-squares optimization. 6.2 Applications: Predictive modeling – House price prediction, Fraud detection; Clustering – Customer segmentation; Use cases in healthcare; Time series – Weather forecasting; Recommendation engines – Product	4

		recommendation; Ethics, bias, and privacy in data science; Reproducible notebooks and version control using Git.	
		Total	39

Textbooks:

1	Steven Skiena, "Data Science Design Manual", Springer International Publishing AG
2	Vijay Kotu, Bala Deshpande. "Data Science Concepts and Practice", Elsevier, M.K. Publishers.
3	Samir Madhavan. "Mastering Python for Data Science", PACKT Publishing
4	Dr. P. N. Arora, Sumeet Arora, S. Arora, Ameet Arora, "Comprehensive Statistical Methods", S.Chand Publications, New Delhi.

Reference Books:

1	Jake VanderPlas. "Python Data Science Handbook", O'reilly Publications.
2	Francesco Ricci, Lior Rokach, Bracha Shapira, Paul B. Kantor, "Recommender Systems Handbook", Springer.
3	S.C. Gupta, V. K. Kapoor "Fundamentals of Mathematical Statistics", S. Chand and Sons, New Delhi.
4	B .L .Agrawal. "Basic Statistics", New Age Publications, Delhi.

Access to software and virtual labs:

1	https://onlinecourses.nptel.ac.in/noc22_cs32/preview
2	https://onlinecourses.nptel.ac.in/noc21_cs69/preview
3	https://www.coursera.org/specializations/applied-data-science
4	www.IntroDataScience.com .
5	https://rapidminer.com/
6	https://towardsdatascience.com/machine-learning/home

Industry articles and case studies :

1	https://www.analyticsvidhya.com/blog/2021/05/data-science-in-healthcare/
2	10 Real World Data Science Case Studies with Examples and Solutions

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3	Data Science in Action: Case Studies (healthcare, finance, retail, etc.)
4	Best Data Science Case Studies of 2025
5	Predictive Analytics Case Studies (enterprise examples)
Any other (Access to AI tools / Data driven insights (if applicable) or any other):	
1	https://h2o.ai/
2	https://datasquirrel.ai/
3	https://flourish.studio/

Internal Assessment:

Internal Assessment consists of one Mid Term Test of 20 marks and Continuous Assessment of 20 marks. The Mid Term test is to be conducted when approximately 50% syllabus is completed and its duration will be one hour.

Continuous Assessment:

Continuous Assessment is of **20 marks**. The rubrics for assessment will be considered on approval by the subject teachers. The rubrics can be any 2 or max 4 of the following table

Sr. No	Rubrics	Marks
1	Certificate course for 4 weeks or more: NPTEL/ Coursera/ Udemy/any MOOC	10 marks
2	Wins in the event/competition/hackathon	10 marks
3	Content beyond syllabus presentation	10 marks
4	Creating Proof of concept	10 marks
5	Mini Project / Extra Experiments/ Virtual Lab	10 marks
6	GATE Based Assignment test/Tutorials etc.	5 marks
7	Participation in event/workshop/talk / competition followed by small report and certificate of participation relevant to the subject (in other institutes)	05 marks
8.	Multiple Choice Questions (Quiz)	05 marks
9.	Peer Review and participation the marks can be left blank (with discretion of faculty)	05 Marks

End Semester Theory Examination:

1	Question paper will be of 60 marks
2	Question paper will have a total of five questions
3	All questions have equal weightage and carry 20 marks each
4	Any three questions out of five need to be solved.

Assessment:

Direct Assessment

End Semester Theory Examination: 60 Marks

1	The question paper will consist of 5 questions, each carrying 20 marks.
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2	The students need to solve any 3 questions.
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Internal Assessment: 40 Marks (Mid Term Test 20 Marks + Continuous Assessment 20 Marks)

The assessment consists of one Mid-team Test of 20 Marks and a Continuous Assessment of 20 Marks. The Mid Term Test is to be conducted when approximately 50% syllabus is completed and its duration will be one hour.

Continuous Assessment: 20 marks

Continuous Assessment is of 20 marks. The rubrics for assessment will be considered upon approval by the subject teachers. It should be a minimum of 2 or a maximum of 4 from the following table

No.	Rubrics	Marks
1	Multiple Choice Questions (Quiz)	5 Marks
2	Participation in event/workshop/talk/competition followed by a small report and certificate of participation relevant to the subject	5 Marks
3	Wins in the event/competition/hackathon pertaining to the course	10 Marks
4	Case study, Presentation, group discussion, technical debate on recent trends in the said course	10 Marks
5	Project-based Learning and evaluation / Extra assignment / Question paper solution	10 Marks
6	NPTEL/ Coursera/ Udemy/any MOOC Certificate course for 4 weeks	10 Marks
7	Content beyond syllabus presentation	10 Marks
8	Mini Project / Extra Experiments/ Virtual Lab	10 Marks
9	GATE Based Assignment tests/Tutorials etc	10 Marks

*For sr.no.6, the date of the certification exam should be within the term, and in case a student is unable to complete the certification, the grading has to be done accordingly.

Indirect Assessment:

1	Mock Test
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2	Skill Enhancement Lecture
3	Extra Assignments/lab/lecture

DATA SCIENCE LAB

Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/PR	Tut	Total
NITPEC711	Data Science Lab	—	02	---	—	01	---	01
Course Code	Course Name	Examination Scheme						
		Theory			Term Work	Practical & Oral	—	Total
		Internal Assessment		End Sem Exam				
		Mid-Term Test	Continuous Assessment					
NITPEC61	Data Science Lab	—	—	—	25	—	—	25

Module wise Suggested Experiments List:

Sr. No.	Module	Experiments	CO Mapping
I	Introduction to Data Science	Data Acquisition and Transformation from Semi-Structured Sources (JSON/API/Logs)	CO1
		Data Wrangling and Pre-processing Using Pandas	
II	Data Exploration (Statistics + EDA)	Statistical Analysis of Dataset Using Descriptive Measures	CO2
		Exploratory Data Analysis and Visualization for Pattern Discovery	
		Hypothesis Testing and Statistical Inference on Real-World Data	
III		Model Validation Using Cross-Validation and Bootstrapping Techniques	CO3

	Methodology and Data Visualization	Data Storytelling Through Advanced Visualizations and Dashboarding	
IV	Anomaly Detection	Outlier Detection Using Statistical Methods (IQR and Z-Score)	CO4
		Anomaly Detection Using Machine Learning Techniques (Isolation Forest and LOF)	
		Handling Imbalanced Data Using SMOTE and Performance Evaluation	
V	Time Series Forecasting	Time Series Decomposition and Stationarity Analysis	CO5
		Time Series Forecasting Using ARIMA Model	
		Comparative Time Series Forecasting Using ARIMA and Prophet	
VI	Optimization and Applications	Least Squares Optimization for Regression Analysis	CO6
		End-to-End Predictive Modeling Case Study (House Price / Fraud Detection)	
Mini Project: Complete Data Science Lifecycle on Real-World Dataset (As a part of continuous assessment)			CO1-6

Note:

- 1) Programs to be implemented in Programming Languages like C ,java, Python.
- 2) Suggested List of Experiments is indicative. However, flexibilities lie with individual course instructor to design and introduce new, innovative and challenging experiments, (limited to maximum 30% variation to the suggested list) from within the curriculum, so that, the fundamentals and applications can be explored to give greater clarity to the students and they can be motivated to think differently.

Term Work:	
1	Term work should consist of 10 experiments and 2 assignments.
2	The final certification and acceptance of term work ensures satisfactory performance of laboratory work and minimum passing marks in term work.
3	Total 25 Marks (Experiments: 15-marks, assignments-5 marks, Attendance - 5 marks)

Practical/Oral Exam: Not Applicable.

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COURSE NAME: SOCIAL MEDIA ANALYTICS

Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/PR	Tut	Total
NITPEC712	Social Media Analytics (Theory)	03	---	---	03	---	---	03

Examination Scheme							
Course Code	Course Name	Theory			Term Work	Practical & Oral	Total
		Internal Assessment		End Sem Exam			
		Mid-Term Test	Continuous Assessment				
NITPEC712	Social Media Analytics (Theory)	20	20	60	---	---	100

Syllabus:

Unit 1:

Social Media Analytics Overview:

- Foundations of Social Media:
- Definition and core characteristics
- Classification of platforms: Social networking, visual sharing, video sharing
- The Social Media Analytics (SMA) Cycle
- Seven Layers of Social Media Analytics
- The 5 V's of Social Data: Volume, Velocity, Variety, Veracity, and Value.
- Structured vs. Unstructured Data: Processing text, images, and videos.
- Engagement Metrics, Conversion & ROI

Unit 2:

Pre-Processing, Social Network Analysis

- Web Page & Social Data Extraction: API-Based Collection, Web Scraping Fundamentals, Web Crawling Strategies\
- Social-Specific Text Cleaning: Noise Removal, Emoji & Emoticon Processing, Handling Slang & Abbreviations

- Text Normalization & Linguistic Preparation: Tokenization, case, Stemming vs. Lemmatization, POS Tagging
- Technical & Ethical Frameworks: Encoding Standards, Duplicate Detection
- Web Crawling: Basic Crawler Algorithm, Crawler Architecture, Traversal Strategies, Specialized Crawler Types

Unit 3:

Social Network Structure and analysis

- Network Anatomy: Nodes and Edges, Graphs, Weighted Networks
- Structural Properties: Connectivity & Cohesion, Density, Network Diameter
- Centrality Measures: Degree Centrality, Betweenness Centrality, Closeness Centrality, Eigenvector Centralit
- Community Detection: Homophily, Modularity, Influencer & Bottleneck Identification
- Visualization Techniques: Layout Algorithms (Fruchterman-Reingold), Color & Size Coding, Dynamic Visualization
- HITS Algorithm

Unit 4:

Social Media Hyperlink Analytics and tools:

- Types of Hyperlinks: **In-links, Out-links, Co-links**
- Methods of Analysis: Hyperlink Environment Analysis, Link Impact Analysis, Social Media Hyperlink Analysis
- Key Metrics & Data Points: Click-Through Rate (CTR), Referral Traffic, Geographic & Behavioural Data
- Link Strategy & Optimization
- Essential Hyperlink & Social Analytics Tools category: Traffic & Referral Tracking, URL Management, All-in-One Social Suites, SEO & Backlink Analysis, Specialized Analysis

Unit 5:

Social Information Filtering, Sentiment Analysis and Opinion Mining

- Social Information Filtering: Information Overload, Automated Recommendation Systems, Traditional vs. Social Recommendations
- Collaborative Filtering (CF) Mechanisms: User-Based, Item-Based, Cold-Start Problem
- Social Network Filtering Strategies: Trust-Based, Social Browsing, Collaborative Tagging & Bookmarking
- Leading Tools for Social Monitoring & Reputation Filtering
- Sentiment Analysis :Levels of Analysis, Core Methodologies, The Sentiment Analysis Process,
- Challenges in sentiment analysis.
- Performance Metrics in Sentiment analysis`
- Opinion Mining: key task, methodology

Unit 6:

- Web Usage Mining
- Predictive & Descriptive Modeling: Deep Cluster Analysis: Beyond basic K-means; exploring Density-Based Spatial Clustering (DBSCAN), Hierarchical Clustering
- Privacy, Ethics & Emerging Trends
- 3. Travel & Hospitality: Hotel Industry Effectiveness, how social media analytics plays a role on it?
- Social media analytics and data mining : "invisible decision maker"

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COURSE NAME: GENERATIVE ARTIFICIAL INTELLIGENCE (GEN AI)

Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/PR	Tut	Total
NITPEC721	Generative AI (Theory)	03	---	---	03	---	---	03

Examination Scheme							
Course Code	Course Name	Theory			Term Work	Practical & Oral	Total
		Internal Assessment		End Sem Exam			
		Mid-Term Test	Continuous Assessment				
NITPEC721	Generative AI (Theory)	20	20	60	---	---	100

Course Prerequisite: Introduction of Python programming language.	
Course Objectives:	
1	Understand the fundamentals of Neural Networks and their different architectures, including RNNs, LSTMs, and Transformers.
2	Learn and explore Generative AI models such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Large Language Models (LLMs).
3	Analyze and compare different AI models used for text, image, and music generation
4	Gain hands-on experience with Python and TensorFlow 2 for implementing and fine-tuning Generative AI models.
5	Evaluate the limitations and ethical concerns related to Generative AI applications in real-world scenarios.
6	Understand the impact of Generative AI in business and industry and explore its potential in various domains like healthcare, finance, and media.
Course Outcomes:	
After successful completion of the course students will be able to:	
1	Explain the evolution of AI and deep learning and their role in Generative AI.

2	Apply different Neural Network architectures for image, text, and music generation.
3	Design and optimize Generative AI models such as GANs and VAEs for real-world problems
4	Analyze the performance of different Generative AI models using evaluation metrics like BLEU, ROUGE, and FID.
5	Demonstrate awareness of ethical and legal implications surrounding Generative AI technologies.
6	Demonstrate AI application in different industry verticals.
Learning Outcomes:	
1	Understand and apply Generative AI fundamentals such as Transformer models, GANs, and VAEs.
2	Develop AI-powered applications for text generation, image synthesis, and music composition.
3	Implement large-scale AI models using deep learning frameworks such as TensorFlow and PyTorch.
4	Critically evaluate AI-generated content for quality, bias, and ethical considerations.
5	Use Generative AI tools from Hugging Face, OpenAI, and other platforms to build AI-driven applications.
6	Apply AI techniques to solve real-world problems in business, healthcare, and other industries.

Syllabus:

Module	Contents	Hrs
Prerequisite	● Introduction to Artificial Intelligence (AI) ○ History and Evolution of AI, Machine Learning (ML), and Deep Learning (DL) ○ Traditional AI vs. Modern AI ○ AI applications in various industries ● Neural Networks and Deep Learning ○ Structure of Artificial Neural Networks (ANNs) ○ Activation Functions: Sigmoid, ReLU, Tanh, Softmax ○ Loss Functions and Optimization (Gradient Descent, Adam)	03

	● Training Neural Networks ○ Backpropagation and Weight Optimization ○ Hyperparameters: Learning Rate, Batch Size, Epochs ○ Overfitting, Underfitting, and Regularization Techniques ● Natural Language Processing (NLP) ○ Basics of NLP: Tokenization, Stemming, Lemmatization ○ Word Embeddings: Word2Vec, GloVe ○ Preprocessing Techniques for Text Data	
Module 1	Introduction to Large Language Models ● What are LLMs? Their evolution and importance ● Use Cases: Chatbots, Text Summarization, Sentiment Analysis Transformer Architecture ● Attention Mechanism in NLP ● Multi-Head Attention and Self-Attention ● Positional Encoding and Tokenization Pre-trained AI Models ● Overview of GPT (Generative Pre-trained Transformer) ● Understanding BERT (Bidirectional Encoder Representations from Transformers) ● Fine-Tuning LLMs for Specific Applications Training and Deployment of LLMs ● Training from Scratch vs. Fine-Tuning ● Model Distillation and Transfer Learning ● Risks and Ethical Considerations in Large Language Models Self-learning Topics: GPT Models and Applications: Applications of GPT models in text generation and dialogue systems. Case study based implementation of GPT-based tasks. GPT-based chatbot enhances E-Shop's customer support service	06
Module 2	What is Generative AI? ● Difference Between Generative and Discriminative Models ● Applications in Text, Image, and Music Generation Generative AI Architectures ● Encoder-Decoder Framework ● Autoencoders and Variational Autoencoders (VAEs) ● Adversarial Training Basics Transfer Learning in Generative AI ● Importance of Pre-trained Models	05

	• Transfer Learning Strategies Ethical and Societal Implications of Generative AI • Bias in AI Models • Privacy and Security Considerations • Deepfake Technology and Misinformation Self-learning Topics: Multinomial Distribution.	
Module 3	• Introduction to Autoencoders ○ Encoder-Decoder Architecture ○ Latent Space Representations ○ Applications of Autoencoders in AI • Variational Autoencoders (VAEs) ○ Understanding Variational Inference ○ The Reparameterization Trick ○ Training VAEs for Image and Text Data • Applications of VAEs ○ Image Reconstruction and Anomaly Detection ○ Text and Speech Synthesis Using VAEs • Limitations of VAEs ○ Trade-off Between Reconstruction Quality and Latent Space Representation ○ Mode Collapse and Performance Bottlenecks Self-learning Topics: Evaluation of VAEs.	06
Module 4	• Introduction to GANs ○ What are GANs? How do they work? ○ Difference Between Generator and Discriminator • Training GANs ○ Loss Functions in GANs (Minimax, Wasserstein Loss) ○ Mode Collapse and Overcoming Training Instabilities • Advanced GAN Architectures ○ StyleGAN, CycleGAN, Pix2Pix ○ Conditional GANs (CGANs) for Controlled Generation • Real-World Applications of GANs ○ Image Synthesis and Deepfake Generation ○ Super-Resolution and Data Augmentation ○ Healthcare and Medical Imaging Applications • Challenges and Limitations of GANs ○ Evaluation Metrics: Inception Score, Fréchet Inception Distance (FID) ○ Stability Issues in Training GANs	05
Module 5	• Understanding Sequence Data ○ Sequential vs. Non-Sequential Data ○ Applications in Text, Speech, and Music Generation • Recurrent Neural Networks (RNNs) ○ Architecture and Functioning of RNNs	07

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	○ Problems with RNNs (Vanishing Gradient) ● Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRUs) ○ Improvements over Traditional RNNs ○ Applications of LSTMs in Text Generation ● Transformer Models and Attention Mechanisms ○ Understanding Attention and Self-Attention ○ How Transformers Overcome RNN Limitations ○ BERT vs. GPT for Text Generation ● Applications in Text, Music, and Creative AI ○ Chatbots and Conversational AI ○ AI-Assisted Music Composition ○ AI-Generated Poetry and Storytelling	
Module 6	Evaluation Metrics for AI-Generated Content ● BLEU Score (Text Generation) ● ROUGE Score (Text Summarization) ● Perplexity Score (NLP Model Evaluation) ● Fréchet Inception Distance (FID) for GANs Use Cases of Generative AI in Industry ● Generative AI for Drug Discovery ● AI for Sustainable Development and Climate Change ● AI for Automated Content Generation Case Studies in Generative AI ● Ethical Issues in Generative AI (Deepfakes, Bias) ● Generative AI for Personalized Marketing ● AI in Legal Document Summarization Future of Generative AI ● Advancements in Zero-Shot and Few-Shot Learning ● AI in Augmented Reality (AR) and Virtual Reality (VR) ● Future Trends and Challenges Self-learning Topics: Responsible AI	07
	Total	39

Textbooks:	
1	"Generative Artificial Intelligence: A Complexity Theory Approach" by David M. Bossens, Bernard H. C. Cheng, and Mohamed O. M. Abdelhafiz
2	"Generative Models" (Online Book) by Shakir Mohamed, Danilo J. Rezende, and Ivo Danihelka
3	"Generative AI - The Frontier of Artificial Intelligence: Unleashing the Power of Imagination" by Kiruthika Subramani.

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4	Generative AI on AWS Building Context-Aware Multimodal Reasoning Applications
Reference Books:	
1	"Hands-On Generative Adversarial Networks with Keras" by Jason Brownlee Focusing on Generative Adversarial Networks (GANs),
2	"Generative AI with Python and TensorFlow 2" by Joseph Babcock, Raghav Bali.
3	Generative AI in Practice by Bernard Marr
4	Applied Generative AI for Beginners (by Akshay Kulkarni, Adarsha Shivananda, Anush kulkarni, Dilip G.
Access to software and virtual labs:	
1	https://cloud.google.com/transform/101-real-world-generative-ai-use-cases-from-industry-leaders
2	
3	
4	
Industry articles and case studies :	
1	
2	
3	
Any other (Access to AI tools / Data driven insights (if applicable) or any other):	
1	https://elearn.nptel.ac.in/shop/iit-workshops/completed/leveraging-generative-ai-for-teaching-programming-courses/?v=c86ee0d9d7ed
2	GitHub Copilot (Limited Access): https://github.com/features/copilot
3	https://newsletter.languagemodels.co/p/the-illustrated-deepseek-r1
4	https://aws.amazon.com/blogs/machine-learning/category/artificial-intelligence/generative-ai/
5	https://lmagents-learning.org
6	https://www.linkedin.com/blog/engineering/generative-ai

Internal Assessment:

1. Assessment consists of one Mid Term Test of 20 marks and Continuous Assessment of 20 marks.
2. Mid Term test is to be conducted when approx. 50% syllabus is completed.
3. Duration of the midterm test shall be one hour.

Continuous Assessment:

Continuous Assessment is of **20 marks**. The rubrics for assessment will be considered on approval by the subject teachers. The rubrics can be any 2 or max 4 of the following:

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Sr. No	Rubrics	Marks
1	Certificate course for 4 weeks or more: NPTEL/ Coursera/ Udemy/any MOOC	10 marks
2	Wins in the event/competition/hackathon	10 marks
3	Content beyond syllabus presentation	10 marks
4	Creating Proof of concept	10 marks
5	Mini Project / Extra Experiments/ Virtual Lab	10 marks
6	GATE Based Assignment test/Tutorials etc	10 marks
7	Participation in event/workshop/talk / competition followed by small report and certificate of participation relevant to the subject (in other institutes)	05 marks
8.	Multiple Choice Questions (Quiz)	05 marks
9.	Peer Review and participation the marks can be left blank (with discretion of faculty)	05 Marks

End Semester Theory Examination:

1	Question paper will be of 60 marks
2	Question paper will have a total of five questions
3	All questions have equal weightage and carry 20 marks each
4	Any three questions out of five needs to be solved.

GENERATIVE ARTIFICIAL INTELLIGENCE (GEN AI) (Lab)

Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Practical	Tut	Total
NITPEC721	Generative AI (Lab)	---	02	---	---	01	---	01

Examination Scheme

Course Code	Course Name	Theory			Term Work	Practical & Oral	Total
		Internal Assessment		End Sem Exam			
		Mid-Term Test	Continuous Assessment				
NITPEC721	Generative AI (Lab)	---	---	---	25	---	25

Suggested Experiments: Students are required to complete at least 10 experiments.	
Star (*) marked experiments are compulsory.	
Sr. No.	Name of the Experiment
1*	Introduction to Python and TensorFlow for Generative AI (*) - Objective: Understand the basics of Python for AI development. - Tasks: - Write Python scripts for basic mathematical operations using NumPy. - Implement matrix operations using TensorFlow. - Explore TensorFlow tensors and data structures. - Implement activation functions: ReLU, Sigmoid, and Tanh..
2*	Data Preprocessing for Generative AI Applications (*) - Objective: Preprocess and clean datasets for AI model training. - Tasks: - Load datasets (e.g., MNIST, CIFAR-10, IMDB Reviews). - Handle missing data, outliers, and normalization. - Apply text tokenization using NLTK and Hugging Face tokenizers. - Convert images to tensors and apply data augmentation.
3*	Data Visualization for AI Models (*) Objective: Analyze datasets visually before training AI models. - Tasks: Generate histograms, box plots, and scatter plots using Matplotlib & Seaborn. Plot word frequency distributions for NLP datasets. Visualize image pixel intensities before model training.
4	Implementing a Generative Adversarial Network (GAN) for Image Generation (*) - Objective: Train a basic GAN model on the MNIST dataset. - Tasks: - Implement a Generator and Discriminator network using TensorFlow/Keras. - Train the GAN model to generate handwritten digits. - Evaluate performance using the Inception Score. - Modify hyperparameters (e.g., batch size, learning rate) to improve results.
5*	Fine-Tuning GAN Models for High-Resolution Image Generation - Objective: Train and optimize GANs for realistic image generation. - Tasks: - Train a DCGAN (Deep Convolutional GAN) for image synthesis.

	○ Implement a Wasserstein GAN (WGAN) to improve stability. ○ Evaluate the quality of generated images using the Fréchet Inception Distance (FID). ○ Experiment with different datasets (e.g., CelebA for face generation).
6*	Variational Autoencoders (VAEs) for Image Generation (*) ● Objective: Train a Variational Autoencoder (VAE) for image reconstruction. ● Tasks: ○ Implement an Encoder-Decoder architecture using TensorFlow. ○ Train a VAE on the MNIST dataset for digit reconstruction. ○ Experiment with different latent space dimensions. ○ Generate new images from sampled latent vectors.
7	Text Generation using Recurrent Neural Networks (RNNs) and LSTMs (*) ● Objective: Generate text using AI models. ● Tasks: ○ Implement a character-level RNN for text generation. ○ Train an LSTM model on a dataset (e.g., Shakespeare text, Wikipedia articles). ○ Generate novel text based on the trained model. ○ Compare generated results with different hyperparameters.
8*	Implementing Transformer-Based Models for Text Generation (GPT & BERT) (*) ● Objective: Fine-tune a pre-trained GPT model for text generation. ● Tasks: ○ Use Hugging Face's Transformers library to load a pre-trained GPT-2 model. ○ Fine-tune the GPT-2 model on a domain-specific dataset. ○ Implement BERT-based text completion and compare results with GPT. ○ Evaluate model performance using BLEU and ROUGE scores.
9	Music Generation using AI (LSTMs & Transformers) (*) ● Objective: Generate new music compositions using AI. ● Tasks: ○ Preprocess MIDI music files for training. ○ Implement an LSTM-based AI music generator. ○ Train a Music Transformer (MusicBERT, MusicGPT) on MIDI sequences. ○ Generate novel music compositions and compare results.
10	AI-Powered Chatbot Development using Large Language Models (LLMs) (*) ● Objective: Create a chatbot using a pre-trained LLM. ● Tasks: ○ Load a pre-trained GPT model for chatbot responses. ○ Fine-tune the model on a customer support dataset.

	○ Implement a chatbot interface using Flask/Streamlit. Deploy the chatbot as a web application.
11	Style Transfer Using Generative AI ● Objective: Apply Neural Style Transfer (NST) to transform images. ● Tasks: ○ Implement a CNN-based style transfer model in TensorFlow. ○ Use VGG19 to extract style and content representations. ○ Apply style transfer to transform real images.
12	Image Inpainting using GANs ● Objective: Fill missing parts of an image using AI. ● Tasks: ○ Implement Context Encoders using GANs. ○ Train the model on CelebA dataset to reconstruct missing facial parts. ○ Evaluate model performance using PSNR and SSIM.
13	Implementing Responsible AI and Bias Mitigation Techniques ● Objective: Address ethical concerns in AI model generation. ● Tasks: ○ Detect bias in AI-generated text. ○ Implement explainability techniques (SHAP, LIME) to interpret model decisions. ○ Evaluate fairness in AI-generated content.
14	Each student must complete a mini-project that involves implementing a Generative AI model for real-world applications. Project Requirements: Project Proposal Submission: ○ The project must focus on text, image, or music generation. ○ Students must choose a real-world dataset and define the problem statement. Implementation Phase: ○ The project must use at least one deep learning framework (TensorFlow/PyTorch). ○ Must include data preprocessing, model training, and evaluation. Evaluation Criteria: ○ Model accuracy and performance (FID, BLEU, ROUGE scores). ○ Innovative approach and complexity. ○ Presentation and documentation

Note: Suggested List of Experiments is indicative. However, flexibilities lie with individual course instructor to design and introduce new, innovative and challenging experiments, (limited to maximum 30% variation to the suggested list) from within the curriculum, so that, the fundamentals and applications can be explored to give greater clarity to the students and they can be motivated to think

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differently.

Term Work:	
1	Term work should consist of 10 experiments.
2	The final certification and acceptance of term work ensures satisfactory performance of laboratory work and minimum passing marks in term work.
3	Total 25 Marks (Experiments: 15-marks, Term work Assessment: 10-marks)

Practical/Oral Exam: 25 Marks Practical/Oral Exam will be held based on above syllabus.

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COURSE NAME: SOFT COMPUTING

Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/PR	Tut	Total
NITPEC722	Soft Computing (Theory)	03	---	---	03	---	---	03

Examination Scheme							
Course Code	Course Name	Theory			Term Work	Practical & Oral	Total
		Internal Assessment		End Sem Exam			
		Mid-Term Test	Continuous Assessment				
NITPEC722	Soft Computing (Theory)	20	20	60	---	---	100

Course Prerequisite: Introduction of C programming language.

Course Objectives:

1	To provide an understanding of soft computing principles and their significance in solving complex real-world problems.
2	To introduce classical neural network models and their role in pattern recognition and efficient computation.
3	To develop an understanding of fuzzy logic and its application in handling uncertainty and enabling interpretable decision-making systems.
4	To present hybrid intelligent systems and their application in designing autonomous and explainable solutions.
5	To introduce evolutionary computation techniques and their use in solving optimization problems.
6	To provide knowledge of edge computing concepts and their integration with soft computing techniques for real-time intelligent systems.

Course Outcomes:

After successful completion of the course students will be able to:

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1	Apply soft computing techniques to solve real-world engineering problems
2	Analyze classical neural network models for pattern recognition and signal processing applications
3	Evaluate fuzzy inference systems for uncertainty handling and explainable decision-making
4	Design neuro-fuzzy systems for intelligent and explainable applications
5	Apply and evaluate genetic algorithms for solving optimization problems
6	Analyze and design intelligent systems using soft computing techniques for edge-based applications
Learning Outcomes:	
1	Students will be able to apply soft computing techniques to solve real-world engineering problems.
2	Students will be able to analyze classical neural network models for pattern recognition and signal processing applications.
3	Students will be able to evaluate fuzzy inference systems for uncertainty handling and explainable decision-making.
4	Students will be able to design neuro-fuzzy systems for intelligent and explainable applications.
5	Students will be able to apply and evaluate genetic algorithms for solving optimization problems.
6	Students will be able to analyze and design intelligent systems using soft computing techniques for edge-based applications.

Module	Contents	Hrs
Prerequisite	Software Engineering	
1	Foundations of Soft Computing: Soft Computing: Definition, Need, Characteristics, Hard vs Soft Computing, Components of Soft Computing, Applications: Healthcare, IoT, Control Systems, Introduction to Hybrid Intelligent Systems	04

2	Neural Network Concepts (Classical & Lightweight): Fundamentals: Biological vs Artificial Neuron, Perceptron Model, Linearly and Non-linearly Separable Problems Learning Concepts (Non-AIML Focus): Hebbian Learning, Competitive Learning, Error-Correction Learning Neural Models: Hopfield Network, Bidirectional Associative Memory (BAM), Radial Basis Function (RBF) Networks Lightweight Neural Models: Concept and Need, Shallow Neural Networks, Extreme Learning Machine (ELM) (conceptual) Applications: Pattern Recognition, Signal Processing, Basics of Medical Imaging.	04
3	Fuzzy Systems, Approximate Reasoning & Explainable AI: Core Concepts: Fuzzy Sets and Membership Functions, Operations on Fuzzy Sets, Fuzzy Relations and Compositions Fuzzy Inference Systems: Mamdani Model, Sugeno Model, Defuzzification Methods Advanced Concepts: Fuzzy Rule-Based Systems, Approximate Reasoning, Handling Uncertainty Explainable AI (XAI) using Fuzzy Systems: Interpretability of fuzzy rule-based systems, Transparent decision-making using IF-THEN rules, Role of fuzzy systems in Explainable AI Applications: Decision Support Systems, Medical Imaging (segmentation, noise handling)	06
4	Hybrid Systems, Neuro-Fuzzy & Agentic AI: Core Concepts: Hybrid Systems: Concept and Need, Neuro-Fuzzy Systems, ANFIS: Architecture (conceptual), Learning mechanism (overview) Agentic AI (Conceptual Integration): Intelligent Agents: Concept and Characteristics, Types of Agents (Reactive, Simple), Fuzzy Rule-Based Agents, Neuro-Fuzzy Agents (conceptual). Explainability in Hybrid Systems: Interpretability vs Accuracy, Role of Neuro-Fuzzy systems in explainable decision-making. Applications: Intelligent decision systems, Healthcare systems.	05
5	Evolutionary Computing & Genetic Algorithms:	04

	Genetic algorithm: Fundamental, basic concepts, working principle, encoding, fitness function, reproduction, Genetic modeling: Inheritance operator, cross over, inversion & deletion, mutation operator, Bitwise operator, Generational Cycle, Convergence of GA, Applications & advances in GA, Differences & similarities between GA & other traditional methods. Multi-objective Optimization (conceptual) :Hybrid GA Systems. Applications: Optimization problems, Feature selection (conceptual), Medical imaging parameter tuning. Self-learning Topics: Real-world case studies of GA in industry (e.g., logistics, finance), Role of GA in Artificial Intelligence and Machine Learning pipelines	
6	Edge Computing for Soft Computing Systems: Introduction: Edge Computing: Definition and Need, Edge vs Cloud Computing, Edge Architecture: Devices, Gateways, Edge Nodes Soft Computing at Edge: Role of Soft Computing in Edge Systems ,Fuzzy logic for real-time decision-making, Lightweight neural models for edge devices, Genetic algorithms for resource optimization. Agent-Based Edge Intelligence: Agents at edge nodes, Autonomous decision-making at edge, Distributed intelligent systems. Applications: Smart Healthcare Systems (real-time monitoring), Medical Imaging at Edge (basic concepts), IoT and Smart Systems Emerging Trends: Edge Intelligence (conceptual), TinyML (overview), Privacy-preserving edge systems Self-learning Topics: Case studies of edge computing in healthcare and smart cities, Energy-efficient model design for edge devices	04
	Total	27

Textbooks:	
1	S.N. Sivanandan and S.N. Deepa, Principles of Soft Computing, Wiley India, 3rd Edition 2018, ISBN: 978-8126577132
2	J.-S. R. Jang, C. –T. Sun, E. Mizutani, Neuro-Fuzzy and Soft Computing, A Computational Approach to Learning and Machine Intelligence, PHI Learning Private Limited RE PRINT-2015.

3	Neural Networks: A Classroom Approach, Satish Kumar, Tata McGraw-Hill Education, 2004/2007
Reference Books:	
1	Anupam Shukla, Ritu Tiwari, Rahul Kala, Real Life Applications of Soft Computing, CRC Press, Taylor & Francis Group, 2010.
2	Genetic Algorithms and Genetic Programming Modern Concepts and Practical Applications © 2009 Michael Affenzeller, Stephan Winkler, Stefan Wagner, and Andreas Beham, CRC Press.
Tools & Platforms:	
1	MATLAB (Fuzzy Logic Toolbox, ANFIS)
2	Python (NumPy, scikit-fuzzy, basic NN libraries)
3	Google Colab (no installation, ideal for students)
4	Edge Impulse (for TinyML & edge demos)
5	Virtual Labs by IIT (simulation-based experiments)
Industry articles and case studies :	
1	https://www.ibm.com/case-studies/edge-computing-healthcare
2	https://towardsdatascience.com/genetic-algorithms-explained-why-they-work-and-how-to-use-them
3	https://ai.google.dev/edge/litert/microcontrollers/overview

Internal Assessment:

- 1) Assessment consists of one Mid Term Test of 20 marks and Continuous Assessment of 20 marks.
- 2) Mid Term test is to be conducted when approx. 50% syllabus is completed.
- 3) Duration of the midterm test shall be one hour.

Continuous Assessment:

Continuous Assessment is of **20 marks**. The rubrics for assessment will be considered on approval by the subject teachers. The rubrics can be any 2 or max 4 of the following:

Sr. No	Rubrics	Marks
1	Certificate course for 4 weeks or more: NPTEL/ Coursera/ Udemy/any MOOC.	10 marks
2	Wins in the event/competition/hackathon.	10 marks

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3	Content beyond syllabus presentation.	10 marks
4	Creating Proof of concept.	10 marks
5	Mini Project / Extra Experiments/ Virtual Lab.	10 marks
6	GATE Based Assignment test/Tutorials etc.	10 marks
7	Participation in event/workshop/talk / competition followed by a small report and certificate of participation relevant to the subject (in other institutes).	05 marks
8.	Multiple Choice Questions (Quiz).	05 marks
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End Semester Theory Examination:

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4	Any three questions out of five need to be solved.

SOFT COMPUTING LAB

Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Practical	Tut	Total
NITPEC722	Soft Computing Lab	---	02	---	---	01	---	01

Examination Scheme

Course Code	Course Name	Theory			Term Work	Practical & Oral	Total
		Internal Assessment		End Sem Exam			
		Mid-Term Test	Continuous Assessment				
NITPEC722	Soft computing Lab	---	---	---	25	25	50

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Suggested Experiments: Students are required to complete at least 10 experiments.	
Star (*) marked experiments are compulsory.	
Sr. No.	Name of the Experiment
1	Implement a simple decision-making system comparing hard vs soft computing approaches for a real-world problem (e.g., temperature control).
2	Analyze the performance of soft computing techniques in handling uncertainty using a sample dataset.
3	Analyze limitations of the perceptron by applying it to a non-linearly separable problem.
4	Compare Hopfield Network and BAM for pattern storage and retrieval performance.
5	Implement a Radial Basis Function (RBF) Network for function approximation or classification.
6	Design a Fuzzy Inference System (Mamdani) for a real-world application (e.g., washing machine control).
7	Evaluate the interpretability of a fuzzy rule-based system for decision-making (Explainable AI perspective).
8	Design and evaluate a Neuro-Fuzzy (ANFIS-based) system for prediction or classification.
9	Implement a Genetic Algorithm for solving an optimization problem (e.g., function optimization or TSP).
10	Analyze the effect of crossover and mutation rates on GA performance.
11	Deploy a lightweight model on edge device/Case study

Note: Suggested List of Experiments is indicative. However, flexibility lies with individual course instructors to design and introduce new, innovative and challenging experiments, (limited to maximum 30% variation to the suggested list) from within the curriculum, so that the fundamentals and applications can be explored to give greater clarity to the students and they can be motivated to think differently.

Term Work:	
1	Term work should consist of 10 experiments.
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